

A MACHINE-CHECKED CONSTRUCTION OF THE UHF ALGEBRA M_{3^∞} AND ITS UNIQUE FAITHFUL TRACE

PABLO COHEN

ABSTRACT. We report a Lean 4 formalization, over mathlib, of the uniformly hyperfinite C^* -algebra M_{3^∞} and its canonical trace. The algebra is built as the metric completion T_∞ of the inductive limit of the matrix tower $M_{3^k}(\mathbb{C})$ under the unital $*$ -embeddings $A \mapsto A \otimes I_3$, and carries mathlib’s `CStarAlgebra` instance. On it we construct a tracial state τ_{UHF} and prove, all kernel-verified, that it is faithful: $\tau_{\text{UHF}}(x^*x) = 0$ implies $x = 0$ for every $x \in T_\infty$. As a corollary T_∞ is simple, every two-sided ideal being $\{0\}$ or the whole algebra, which is Glimm’s 1960 theorem for this algebra. We further prove that τ_{UHF} is the unique continuous unital tracial functional on T_∞ . Faithfulness, simplicity, and uniqueness of the trace together identify T_∞ as the Glimm 3^∞ factor. To the best of our knowledge this is the first machine-checked construction of an infinite-dimensional simple C^* -algebra with a canonical faithful and unique trace. Every headline theorem depends only on the three standard kernel axioms `[propext, Classical.choice, Quot.sound]`, with no project-level axioms and no `sorry`, verified at pinned toolchain and mathlib revisions.

1. INTRODUCTION

The uniformly hyperfinite (UHF) C^* -algebras are among the oldest and best-understood objects in operator algebra theory. They are the norm-closures of increasing unions of full matrix algebras. Glimm classified them in 1960 by a supernatural number and proved that each one is simple and carries a unique tracial state, necessarily faithful [5]. The standard references treat all of this as foundational [3, 1, 12]. The algebra of the present paper is M_{3^∞} : the tower $M_{3^k}(\mathbb{C})$ with the embeddings $A \mapsto A \otimes I_3$, completed in norm.

As a formalization target the object is attractive precisely because it is concrete. It is a single, explicitly presented C^* -algebra whose three defining structural facts, faithfulness of the canonical trace, simplicity, and uniqueness of the trace, are theorems rather than definitions, and each of the three requires a genuine analytic or algebraic argument to establish. A proof assistant either has these theorems or it does not. Before this work no proof assistant did. Formalized operator algebra is young; mathlib, the Lean 4 mathematical library [7, 8], provides a C^* -algebra typeclass hierarchy, an L^2 -operator-norm structure on matrices, ring direct limits, metric completions, and a continuous functional calculus, following the functional-analysis program of Dupuis, Lewis, and Macbeth [4]. It contains no concrete infinite-dimensional simple C^* -algebra, no tracial state theory, no conditional expectations, and no completely positive maps. Building M_{3^∞} and proving the classical facts therefore means assembling, from mathlib’s primitives, a body of infrastructure that the pen-and-paper literature takes for granted, and at several points finding proofs different from the classical ones.

Why faithfulness is the hard property. Everything about τ_{UHF} except faithfulness transfers from the matrix level to the completion by soft means. Additivity, traciality, Hermitian symmetry, and positivity are all closed conditions: each says that a continuous function vanishes, or is nonnegative, on T_∞ , so it is enough to check it on the dense image of the direct limit and pass to the closure. Faithfulness is different in kind. The statement $\tau_{\text{UHF}}(x^*x) = 0 \Rightarrow x = 0$ is an implication with a nontrivial conclusion, and the set of x satisfying it is not the preimage of a

closed set. A density argument cannot reach it: a limit of matrices, each detected by its own trace, could in principle converge to a nonzero element whose trace-of-square vanishes. Ruling this out needs global structure. Classically one has two routes. Either faithfulness comes from simplicity, because the left kernel of a state generates a closed two-sided ideal that must be trivial when none exists, with simplicity supplied by Glimm’s theorem; or it comes from the norm-one conditional expectations onto the finite levels, in the tradition of Tomiyama [13], whose contractivity is classically a consequence of complete positivity and the Kadison–Schwarz inequality [6]. Neither route was available in `mathlib`.

How the proof was actually found. The development did not proceed cleanly, and the record is worth stating. An early version of this material carried a completion-level “faithfulness” declaration that, on inspection, was a placeholder: a named proposition equal to `True`, discharged by `trivial`, together with a theorem whose hypothesis was the conclusion restated. Neither proves anything, and a companion note demoted completion-level faithfulness from conditional to open before the argument was rebuilt.

The first serious attempt reduced faithfulness, in kernel-verified form, to a single hypothesis: that every nonzero two-sided ideal of T_∞ contains an element of nonzero trace. This looked like progress. A four-fold equivalence then showed that the hypothesis is not weaker than the goal but logically equivalent to it, and to simplicity as well. The reduction had restated the summit, not descended from it. The equivalence was worth formalizing anyway, because it is an exact statement of where the classical hard input has to enter, and it ruled out any attempt to reach the result from something strictly softer.

What eventually closed the argument was the conditional-expectation route, built from scratch with elementary replacements for the missing theory. The trace 2-norm contraction of the level- k expectation, classically a corollary of Kadison–Schwarz for a completely positive map, reduced to a three-line finite Cauchy–Schwarz once the target inequality was seen to be scalar rather than operator-valued. The operator-norm contraction, classically the statement that a unital completely positive map has norm one, followed from the isometry decomposition $\mathcal{E}B = \frac{1}{3} \sum_t W_t^* B W_t$ together with the C^* -identity and submultiplicativity, with no complete-positivity theory at all. With both contractions in hand, faithfulness was proved directly, and simplicity was obtained as a corollary through the equivalence above. This inverts the classical order, in which simplicity is proved first and faithfulness deduced from it.

A note on names. The formalization grew out of a larger exploratory project, and the results presented here are self-contained and depend on none of that project’s conjectural content. Lean identifiers retain their original names: the direct limit is `TimelessFieldRing` and its completion `TimelessFieldCompletion` in the source, and many declarations carry a `substrate` or version prefix. These are names, nothing more. In this paper the objects are written $\varinjlim_k A_k$ and T_∞ , and every property claimed of them is one of the kernel-verified theorems named below or a standard cited fact.

Roadmap. Section 2 constructs the tower, its inductive limit, and the completion, and registers the C^* -algebra instance. Section 3 builds τ_{UHF} and its soft properties. Section 4 develops the algebraic engine: a Cauchy–Schwarz inequality for the trace pairing, the trace-null ideal, the Weyl averaging computation, simplicity below the completion, and the four-fold equivalence. Section 5 is the analytic core: the level expectation, its two contraction estimates, the extension to the completion, and the proof of faithfulness, followed by simplicity, uniqueness of the trace, and the identification of T_∞ as a factor. Section 6 records the toolchain and verification data, and Section 7 discusses related work.

2. THE TOWER, THE LIMIT, AND THE COMPLETION

At level $k \in \mathbb{N}$ set

$$A_k := \text{Matrix}(\text{Fin } 3^k) (\text{Fin } 3^k) \mathbb{C},$$

a finite-dimensional C^* -algebra under the L^2 -operator norm. This is mathlib's scoped instance `Matrix.Norms.L2Operator`, which supplies the C^* -identity $\|M^*M\| = \|M\|^2$. Successive levels embed by a ternary tensor construction,

$$\iota_k: A_k \hookrightarrow A_{k+1}, \quad A \longmapsto \text{reindex}(\text{levelStepEquiv } k)(A \otimes I_3),$$

where `levelStepEquiv` k is the equivalence $\text{Fin } 3^k \times \text{Fin } 3 \simeq \text{Fin } 3^{k+1}$ coming from the ternary decomposition of 3^{k+1} , and $\otimes$ is the Kronecker product. In the source this map is `substrateEmbedMatrix`, with iterated form `substrateRingHomIter`. Each ι_k is a unital $*$ -ring homomorphism and an isometry.

Applying mathlib's `Ring.DirectLimit` to $\{A_k, \iota_k\}$ produces the $\mathbb{C}$ -algebra $R := \varinjlim_k A_k$, with a `StarRing` structure descending from the entrywise conjugate transpose. Every element of R has a single-level representative: mathlib's `DirectLimit.exists_eq_mk` produces, from $x \in R$, a level k and a matrix $A \in A_k$ with $x = \llbracket(k, A)\rrbracket$. Because the embeddings are isometric, the level norm descends to a well-defined norm on R that makes it a normed $*$ -ring; the canonical map $\iota_k: A_k \rightarrow R$ is an isometry, and the cocone identity holds up to `Quotient.sound`.

The metric completion of R under this norm is

$$T_\infty := \overline{\varinjlim_k A_k}^{\|\cdot\|_{\text{op}}},$$

built with `UniformSpace.Completion`. The algebraic operations extend by uniform continuity, and the C^* -identity passes to the completion by the usual Cauchy-sequence argument, run against mathlib's completion API.

Theorem 2.1 (UHF C^* -algebra registration). *T_∞ carries mathlib-native `Ring`, `StarRing`, `NormedRing`, and `CStarAlgebra` instances, with $\varinjlim_k A_k$ dense in T_∞ .*

By construction T_∞ is a UHF algebra in the classical sense: the norm-closure of an increasing union of full matrix algebras under unital $*$ -embeddings, of supernatural number 3^∞ . The mathlib `CStarAlgebra` instance is the algebra of record; no appeal is made to the classical classification. Each level is itself simple, via `DivisionRing.isSimpleRing` on $\mathbb{C}$ composed with `IsSimpleRing.matrix`, and this finite-dimensional fact is the seed of the simplicity argument in Section 4. The base 3 plays no special role; every argument below works verbatim for any base ≥ 2 , and the choice is inherited from the source project's ternary conventions.

3. THE TRACE AND ITS SOFT PROPERTIES

On level k the normalized trace $\tau_n(M) := \text{trace}(M)/n$, with $n = 3^k$, is linear, unital, and 1-Lipschitz for the operator norm; the Lipschitz bound uses the Hilbert–Schmidt estimate $\|M\|_{\text{HS}}^2 \leq n\|M\|_{\text{op}}^2$ and Cauchy–Schwarz. The traces are compatible with the embeddings, $\tau_{3^{k+1}}(\iota_k A) = \tau_{3^k}(A)$, by `Matrix.trace_kronecker`. The universal property of the direct limit (`DirectLimit.lift`) yields a pre-trace $\tau_{\text{pre}}: R \rightarrow \mathbb{C}$ (`substrate_pre_trace`) that is additive, unital, and 1-Lipschitz, hence uniformly continuous, and

$$\tau_{\text{UHF}} := \text{UniformSpace.Completion.extension}(\tau_{\text{pre}}): T_\infty \rightarrow \mathbb{C}$$

is its unique continuous extension, agreeing with τ_{pre} on the dense image.

Theorem 3.1 (The canonical tracial state). *$\tau_{\text{UHF}}: T_\infty \rightarrow \mathbb{C}$ is additive, unital, 1-Lipschitz, and uniformly continuous; $\mathbb{C}$ -linear (UHF_trace_smul, bundled as UHF_trace_linearMap); tracial, $\tau_{\text{UHF}}(xy) = \tau_{\text{UHF}}(yx)$ (UHF_trace_mul_comm); Hermitian, $\tau_{\text{UHF}}(x^*) = \overline{\tau_{\text{UHF}}(x)}$; and positive,*

$0 \leq \tau_{\text{UHF}}(x^*x)$ in the complex order (`UHF_trace_star_mul_self_nonneg`, under `mathlib`'s `scopedComplexOrder`).

Each property is a closed condition and follows the same three steps: a `mathlib` matrix lemma at level k (among `Matrix.trace_mul_comm`, `Matrix.trace_conjTranspose`, and `Matrix.posSemidef_conjTranspose_mul_self` with `Matrix.PosSemidef.trace_nonneg`); reduction to a single-level representative; and completion induction (`UniformSpace.Completion.induction_on`, or its binary form for traciality) over the closed set where the identity holds. Closedness of the positivity set uses `Complex.orderClosedTopology`.

Faithfulness stratifies over the three tiers of the construction. At the matrix level, $\tau_n(M^*M) = 0 \Rightarrow M = 0$ is `Matrix.trace_conjTranspose_mul_self_eq_zero_iff`: the trace of M^*M is $\sum_{i,j} |M_{ij}|^2$. At the direct limit, reduce to a single-level representative, apply the matrix case, and push the zero back through the isometric embedding (`substrate_pre_trace_star_mul_self_faithful`). The completion tier, $\tau_{\text{UHF}}(x^*x) = 0 \Rightarrow x = 0$ for $x \in T_\infty$, is the hard case for the reason given in the introduction: it is not a closed condition, and no density argument reaches it. Its proof occupies Sections 4 and 5.

4. THE SIMPLICITY ENGINE AND THE FALSE SUMMIT

This section builds the algebraic machinery that first reduced completion-tier faithfulness to a single hypothesis and then identified that hypothesis as equivalent to the goal. All results are unconditional kernel-verified theorems.

4.1. Cauchy–Schwarz for the trace pairing. Define $\langle x, y \rangle_\tau := \tau_{\text{UHF}}(y^*x)$ on T_∞ (`traceInner`). Positivity (Theorem 3.1) makes this a positive semidefinite sesquilinear form, and the discriminant argument goes through against `mathlib`'s `discrim_le_zero`.

Theorem 4.1 (`UHF_trace_cauchySchwarz`). *For all $x, y \in T_\infty$, $\|\tau_{\text{UHF}}(y^*x)\|^2 \leq \text{Re } \tau_{\text{UHF}}(x^*x) \cdot \text{Re } \tau_{\text{UHF}}(y^*y)$.*

The proof expands $\tau_{\text{UHF}}((x - cy)^*(x - cy)) \geq 0$ along real multiples $c = t \tau_{\text{UHF}}(y^*x)$, using linearity and the `StarModule` structure, and applies the quadratic discriminant bound. One corollary is left absorption of the trace-null cone (`UHF_trace_null_left_absorption`): if $\tau_{\text{UHF}}(x^*x) = 0$ then $\tau_{\text{UHF}}((ax)^*(ax)) = 0$ for every a , since the left side is $\langle a^*ax, x \rangle_\tau$, dominated through Theorem 4.1 by $\text{Re } \tau_{\text{UHF}}(x^*x) = 0$.

4.2. The trace-null two-sided ideal. Let $N := \{x \in T_\infty : \tau_{\text{UHF}}(x^*x) = 0\}$. Theorem 4.1 makes N additive; traciality makes it $*$ -closed, since $\tau_{\text{UHF}}(xx^*) = \tau_{\text{UHF}}(x^*x)$; left absorption and $*$ -closure together give right absorption; scalar closure is immediate.

Theorem 4.2 (`traceNullTwoSidedIdeal`). *N is the carrier of a `TwoSidedIdeal` of T_∞ , and it is proper, since $\tau_{\text{UHF}}(1) = 1$ gives $1 \notin N$ (`traceNullTwoSidedIdeal_ne_top`).*

This yields at once the conditional form of faithfulness, which was the honest theorem available at that stage.

Theorem 4.3 (`UHF_trace_faithful_of_simple`). *If every two-sided ideal of T_∞ is $\{0\}$ or T_∞ , then τ_{UHF} is faithful.*

Indeed N is a proper two-sided ideal, so simplicity forces $N = \{0\}$, and $x \in N$ means $\tau_{\text{UHF}}(x^*x) = 0$, whence $x = 0$. The hypothesis is genuine algebraic simplicity, classically Glimm's theorem.

4.3. The Weyl averaging computation. The engine of every simplicity proof for matrix towers is finite-dimensional. Fix $n \geq 1$, let $\omega := e^{2\pi i/n}$ (a primitive root, `substrateRoot`), and let

$$C := \text{diag}(j \mapsto \omega^j), \quad S_{ij} := [i = j + 1]$$

be the clock and shift matrices [14], both unitary, with powers $C^m = \text{diag}(j \mapsto \omega^{jm})$ and $(S^m)_{ij} = [i = j + m]$. The character sum $\sum_{a < n} (\omega^i \overline{\omega^j})^a = n [i = j]$ (`substrate_clock_character_sum`, via `geom_sum_mul` and `IsPrimitiveRoot.pow_inj`) drives a two-stage computation. Averaging over the clock powers kills the off-diagonal, $\frac{1}{n} \sum_a C^a x (C^a)^* = \text{diag}(\text{diag } x)$; averaging a diagonal over the shift powers equalizes it, $\frac{1}{n} \sum_b S^b D (S^b)^* = (\text{trace } D/n) \cdot 1$. Together:

Theorem 4.4 (`matrix_unitary_average_eq_trace_smul_one`). *For every $x \in M_n(\mathbb{C})$,*

$$\frac{1}{n^2} \sum_{a < n} \sum_{b < n} (C^a S^b) x (C^a S^b)^* = \frac{\text{trace } x}{n} \cdot 1.$$

A set closed under addition, arbitrary two-sided multiplication, and $\mathbb{C}$ -scaling absorbs each conjugate uxu^* , hence absorbs the average: any such absorbing set containing x contains $(\text{trace } x/n) \cdot 1$.

4.4. Simplicity below the completion. Firing the engine at a nonzero element needs a nonzero trace, which a nonzero ideal element need not have. The classical fix is the matrix-unit trick: if $x \neq 0$, pick a nonzero entry x_{pq} ; the matrix unit $E_{q,p}$ satisfies $\text{trace}(E_{q,p}x) = x_{pq} \neq 0$ (`trace_single_mul`), and $E_{q,p}x$ remains in the absorbing set. Averaging and rescaling put 1 into the set. With level simplicity this gives matrix simplicity in ideal form (`matrix_twoSidedIdeal_bot_or_top`), and pulling a two-sided ideal of R back to each level (`substrateIdealPullback`) lifts it to the limit.

Theorem 4.5 (`directLimit_twoSidedIdeal_bot_or_top`). *Every two-sided ideal of $\varinjlim_k M_{3^k}(\mathbb{C})$ is $\{0\}$ or the whole ring.*

The same Weyl words push forward to unitaries $u_{a,b}$ in T_∞ , and the completion-level averaging operator $\text{avgK}(x) := (3^k)^{-2} \sum_{a,b < 3^k} u_{a,b} x u_{a,b}^*$ agrees with $\tau_{\text{UHF}}(\cdot) \cdot 1$ on the image of level k , by Theorem 4.4. Combining density, non-expansiveness of averaging, and the 1-Lipschitz property of τ_{UHF} gives, for every x and $\varepsilon > 0$, a level k with $\|\text{avgK}(x) - \tau_{\text{UHF}}(x) \cdot 1\| < \varepsilon$ (`avgK_approx`). Suppose a two-sided ideal I contains an x with $c := \tau_{\text{UHF}}(x) \neq 0$. Averaging is by conjugation, so $\text{avgK}(x) \in I$; taking $\varepsilon = \|c\|$ gives $\|1 - c^{-1} \text{avgK}(x)\| < 1$, so $c^{-1} \text{avgK}(x)$ is invertible by the Neumann series. An invertible element of an ideal forces $1 \in I$: multiply it by its inverse, which lies in the whole algebra, and the product stays in I because I absorbs multiplication on both sides. No closure of I enters: $\text{avgK}(x)$ is a finite average, a genuine element of I , so the argument applies to algebraic, not merely closed, two-sided ideals. Thus T_∞ is simple, and τ_{UHF} is faithful, provided every nonzero ideal contains an element of nonzero trace.

4.5. The four-fold equivalence. The natural next question is whether that trace-detection hypothesis can be discharged directly. The formalized answer is no: the hypothesis is logically equivalent to the goal it was meant to unlock.

Theorem 4.6 (`traceDetects_tfae`). *For T_∞ the following are equivalent: (1) every two-sided ideal $I \neq \{0\}$ contains an x with $\tau_{\text{UHF}}(x) \neq 0$; (2) $\text{traceNullTwoSidedIdeal} = \{0\}$; (3) τ_{UHF} is faithful; (4) every two-sided ideal of T_∞ is $\{0\}$ or T_∞ . The cycle $(4) \Rightarrow (3) \Rightarrow (2) \Rightarrow (1) \Rightarrow (4)$ is formalized with `mathlib's List.TFAE`.*

This is the honest obstruction map. No hypothesis strictly weaker than faithfulness closes the loop, and the classical hard input, Glimm's theorem or something of equal strength, is genuinely required. The same development also probed the spectral-projection route, which manufactures

a nonzero projection inside an ideal from a nonzero selfadjoint element through the continuous functional calculus, and located exactly where it stalls in the non-unital setting. The route that succeeded is the conditional expectation, and it is different from both.

5. THE LEVEL EXPECTATION AND THE FAITHFULNESS PROOF

The closing argument constructs the canonical projection onto each finite level, proves two contraction estimates for it by elementary means, extends it to the completion, and reads off faithfulness from the reconstruction property. Simplicity then follows through Theorem 4.6. We write τ_{3^k} for the normalized trace on $A_k = M_{3^k}(\mathbb{C})$.

5.1. The partial trace. Under $M_{3^{k+1}}(\mathbb{C}) \cong M_{3^k}(\mathbb{C}) \otimes M_3(\mathbb{C})$, given by `levelStepEquiv`, the single-step partial trace $\text{id} \otimes \tau_3: A_{k+1} \rightarrow A_k$ is, entrywise,

$$(\mathcal{E}B)_{ij} = \frac{1}{3} \sum_{t=0}^2 B_{(i,t),(j,t)} \quad (\text{partialTraceStep}),$$

where (i, t) abbreviates `levelStepEquiv(i, t)`. Iterating gives `partialTraceDown`: $A_j \rightarrow A_k$ for $k \leq j$.

These maps are compatible with the inductive system: for $k \leq j \leq j'$, the iterated partial trace $A_{j'} \rightarrow A_k$ factors through A_j , and $\mathcal{E}$ inverts a single embedding ($\mathcal{E}(A \otimes I_3) = A$, `partialTraceStep_substrateEmbedMatrix`). Consequently the level maps agree on the image of every embedding, so they descend, by the universal property of the direct limit, to a single well-defined map $E_k: \varinjlim_m A_m \rightarrow M_{3^k}(\mathbb{C})$ (`condExp`); concretely, $E_k[(j, A)] = \text{partialTraceDown}(A)$ is independent of the chosen single-level representative. This quotient-invariance is the one nontrivial well-definedness check behind Theorem 5.1.

We call E_k the *level- k expectation map*, and we use only the retraction and trace-preservation properties recorded in Theorem 5.1 below. The full conditional-expectation API, meaning positivity, the A_k -bimodule identity $E_k(axb) = a E_k(x) b$ for $a, b \in A_k$, and idempotence as a norm-one projection, is not formalized. The phrase “conditional expectation” is here shorthand for this partial-trace retraction, not a claim that the classical axioms are machine-checked.

Theorem 5.1 (Properties of E_k). *E_k is additive and $\mathbb{C}$ -homogeneous, unital, star-preserving, inverts the embedding on the level- k corner (`condExp_level_k`, with `condExp_lower_level` for levels $m \leq k$), and is trace-preserving: $\tau_{3^k}(E_k x) = \tau_{\text{pre}}(x)$ (`condExp_normalized_trace`).*

5.2. The 2-norm contraction. The first analytic requirement is contraction of E_k in the trace 2-seminorm:

$$\text{Re } \tau_{3^k}((E_k x)^*(E_k x)) \leq \text{Re } \tau_{\text{pre}}(x^* x) \quad (\text{condExp_l2_contraction}). \quad (1)$$

Classically this is Kadison–Schwarz for the completely positive map E_k , composed with the trace. Mathlib has neither, and formalizing them would have been a project in itself. The formalization targets the scalar inequality directly, and it is elementary. Finite Cauchy–Schwarz over three terms gives $|\sum_{t=0}^2 a_t|^2 \leq 3 \sum_{t=0}^2 |a_t|^2$ (`normSq_sum_fin3_le`), so each entry obeys $|(\mathcal{E}B)_{ij}|^2 \leq \frac{1}{3} \sum_t |B_{(i,t),(j,t)}|^2$. Summing over i, j , the right side is exactly the diagonal-tail sub-sum ($s = t$) of

$$\tau_{3^{k+1}}(B^* B) = \frac{1}{3^{k+1}} \sum_{i,s,j,t} |B_{(j,t),(i,s)}|^2,$$

every term of which is nonnegative. Mathlib’s `Finset.single_le_sum` closes the single-step bound $\tau_{3^k}((\mathcal{E}B)^*(\mathcal{E}B)) \leq \tau_{3^{k+1}}(B^* B)$ (`partialTraceStep_normSq_trace_le`); iterating and descending to the limit gives (1). No positive-semidefinite operator theory and no complete positivity enter. For trace-compatible partial traces the scalar 2-norm contraction is a three-line fact, and the Kadison–Schwarz detour of the textbook proof is avoidable.

5.3. The operator-norm contraction. Extending E_k to the completion requires the operator-norm bound $\|E_k x\| \leq \|x\|$. Classically this is the statement that a unital completely positive map has norm one. The formalized proof uses an isometry decomposition. For $t \in \{0, 1, 2\}$ let $W_t: \mathbb{C}^{3^k} \rightarrow \mathbb{C}^{3^{k+1}}$ be the coordinate injection $v \mapsto v \otimes e_t$, viewed as a $3^{k+1} \times 3^k$ matrix (`Winj`). Then

$$\mathcal{E}B = \frac{1}{3} \sum_{t=0}^2 W_t^* B W_t \quad (\text{partialTraceStep_eq_smul_sum}), \quad (2)$$

and each W_t satisfies $W_t^* W_t = I$. The C^* -identity for the L^2 -operator norm gives $\|W_t\|^2 = \|W_t^* W_t\| = \|I\| \leq 1$ (`Winj_opNorm_sq_le_one`), so submultiplicativity and $\|W_t^*\| = \|W_t\|$ yield $\|W_t^* B W_t\| \leq \|B\|$, and averaging over the three slots gives $\|\mathcal{E}B\| \leq \|B\|$ (`partialTraceStep_opNorm_le`). Iterating gives $\|E_k x\| \leq \|x\|$ (`condExp_opNorm_le`), so E_k is 1-Lipschitz (`condExp_1lipschitz`), hence uniformly continuous. The only inputs are four L^2 -operator-norm lemmas and the norm bound on the identity operator. Decomposition (2) is the concrete Stinespring form of a partial trace [11].

5.4. Extension and the closing argument. Uniform continuity extends E_k to the completion in two forms: the matrix-valued `condExpFin` $k: T_\infty \rightarrow M_{3^k}(\mathbb{C})$, and its recoordinatization `condExpCompletion` $k: T_\infty \rightarrow T_\infty$ given by $x \mapsto \iota_k(\text{condExpFin } k x)$. Both are 1-Lipschitz and agree with E_k on the dense image. Three statements complete the proof.

Lemma 5.2 (Reconstruction, `condExpCompletion_tendsto`). *For every $x \in T_\infty$, $\text{condExpCompletion } k x \rightarrow x$ as $k \rightarrow \infty$.*

The proof is an $\varepsilon/2$ density lift over the uniformly Lipschitz family: approximate x within $\varepsilon/2$ by a dense-image element with representative at level m ; for $k \geq m$ the map fixes that element (`condExpCompletion_fixes_coe_lower`), and the Lipschitz bound controls the difference at x .

Lemma 5.3 (Lifted contraction, `condExpCompletion_l2_contraction`). *For every k and $x \in T_\infty$, $\text{Re } \tau_{\text{UHF}}((\text{condExpCompletion } k x)^*(\text{condExpCompletion } k x)) \leq \text{Re } \tau_{\text{UHF}}(x^* x)$.*

This is an inequality between two continuous real-valued functions of x . It holds on the dense image by (1) and the trace bridge $\tau_{\text{UHF}}((E_k x)^*(E_k x)) = \tau_{3^k}((\text{condExpFin } k x)^*(\text{condExpFin } k x))$, and completion induction on the closed set $\{x : f(x) \leq g(x)\}$ (`isClosed_le`) transfers it to all of T_∞ .

Lemma 5.4 (Matrix faithfulness, real-part form, `matrix_faithful_of_re_nonpos`). *If $\text{Re } \tau_n(M^* M) \leq 0$ then $M = 0$.*

Positivity pins $\tau_n(M^* M)$ to the nonnegative reals; a nonpositive real part then forces it to 0, and matrix-tier faithfulness concludes.

Theorem 5.5 (Faithfulness, `UHF_trace_faithful`). *For every $x \in T_\infty$, if $\tau_{\text{UHF}}(x^* x) = 0$ then $x = 0$. Kernel axioms [`propext`, `Classical.choice`, `Quot.sound`]; zero project axioms.*

Proof. Suppose $\tau_{\text{UHF}}(x^* x) = 0$. By Lemma 5.3 and the trace bridge, $\text{Re } \tau_{3^k}((\text{condExpFin } k x)^*(\text{condExpFin } k x)) \leq 0$ for every k ; by Lemma 5.4, $\text{condExpFin } k x = 0$, hence $\text{condExpCompletion } k x = 0$, for every k . By Lemma 5.2 the constant-zero sequence $\text{condExpCompletion } k x$ converges to x , so uniqueness of limits gives $x = 0$. $\square$

Corollary 5.6 (Simplicity, `substrate_completion_simple_unconditional`). *Every two-sided ideal of T_∞ is $\{0\}$ or T_∞ .*

Proof. Theorem 5.5 is item (3) of Theorem 4.6, and item (4) follows. $\square$

Remark 5.7. The order of deduction is the reverse of Glimm’s: he proves simplicity first and derives faithfulness of the trace, whereas here faithfulness is proved directly through the level expectations and simplicity follows. Both directions of the equivalence are formalized, so Glimm’s implication, Theorem 4.3, is machine-checked too. Corollary 5.6 is the stronger algebraic statement, covering all two-sided ideals rather than only closed ones, which holds for every UHF algebra [3].

5.5. Uniqueness of the trace. The same level-expectation machinery gives the third leg of the identification: τ_{UHF} is the only trace. We package the hypotheses as a predicate `IsTracialState` φ requiring φ to be continuous, additive, $\mathbb{C}$ -homogeneous, unital, and tracial; positivity is deliberately omitted. (The Lean name retains “State” for continuity with the source; because positivity is dropped, the predicate is really that of a continuous unital tracial *functional*, and the theorem is correspondingly stronger than uniqueness among states.)

Theorem 5.8 (Uniqueness, `substrate_UHF_trace_unique`). *For every φ with `IsTracialState` φ and every $x \in T_\infty$, $\varphi(x) = \tau_{\text{UHF}}(x)$. Kernel axioms [`propext`, `Classical.choice`, `Quot.sound`].*

The proof is the standard argument in three steps. At a finite level, any additive, $\mathbb{C}$ -homogeneous, unital, tracial functional on $M_n(\mathbb{C})$ equals the normalized trace (`matrix_tracial_state_unique`): the off-diagonal matrix units are commutators, $E_{ij} = [E_{ik}, E_{kj}]$, hence annihilated, and the diagonal units are cyclic-conjugate, hence equal, so φ is $\varphi(1)/n$ times the trace, and unitality fixes the constant. Mathlib does not carry this lemma, so it is proved from `Matrix.single` identities and `matrix_eq_sum_single`. For $x = \iota_k(A)$, the functional $\varphi \circ \iota_k$ is of this kind, so $\varphi(x) = \tau_{3^k}(A) = \tau_{\text{UHF}}(x)$. Finally φ and τ_{UHF} are continuous and agree on the dense image, so they agree everywhere, by `isClosed_eq`.

Theorem 5.5, Corollary 5.6, and Theorem 5.8 together record, in the capstone `r113_substrate_UHF_factor_capstone`, that T_∞ is a faithful, simple, uniquely-traced UHF C^* -algebra: the Glimm 3^∞ factor.

5.6. Mathlib infrastructure and gaps. The construction is mathlib-native. The load-bearing APIs are `Ring.DirectLimit` with `DirectLimit.lift` and `DirectLimit.exists_eq_mk`; `UniformSpace.Completion` with its `extension`, `induction_on`, and isometry lemmas; the scoped L^2 -operator-norm C^* -structure on matrices; the matrix trace and positivity lemmas; `IsSimpleRing.matrix` and `TwoSidedIdeal`; the roots-of-unity API for the Weyl computation; and the scoped `ComplexOrder` with `discrim_le_zero`.

Three absences shaped the proofs and are natural next targets for mathlib. There is no complete-positivity theory: no Kadison–Schwarz inequality, no partial trace, no “unital CP map has norm one.” This forced, and in the end rewarded, the elementary routes of Sections 5.2 and 5.3, which are shorter than the classical arguments and may be of expository interest on their own. There are no spectral projections: the Borel functional calculus is absent, and the continuous functional calculus produces projections only from clopen spectral sets; the probe of Section 4.5 built exactly that and identified the missing hypothesis. And a `UniformSpace.Completion` does not automatically carry the order structure of a C^* -algebra, so positivity of τ_{UHF} was proved scalar-side in the `ComplexOrder` on $\mathbb{C}$ rather than operator-side. Three standalone files written to mathlib contribution standards were produced during the spectral-projection probe, on topological closures of two-sided ideals, on the non-unital continuous functional calculus mapping into closed ideals, and on spectral projections from clopen quasispectral sets, and are offered upstream.

6. REPRODUCIBILITY

Pinned toolchain. Lean toolchain `leanprover/lean4:v4.24.0-rc1`; mathlib commit `eed770a434957369c6262aa3fb1d6426419016d4`; verified HEAD `91cc1a01` (2026-07-23), with build and `#print axioms` run the same day. The source is public at github.com/FractalDevTeam/Principia-Fractalis; the commit above is a stable reference point in that history.

Build. From the `PF_Lean4_Code` directory, `elaninstall$(catlean-toolchain)` followed by `lakebuildPF` (4,472 jobs) and `lakebuild` (8,930 jobs; the second pass re-elaborates every theorem through the production Lean 4 kernel under a distinct package boundary).

Axiom audit. The audit blocks at the ends of the source files report, for each headline theorem:

```
'PrincipiaTractalis.SubstrateCompletionFaithful.UHF_trace_faithful'
  depends on axioms: [propext, Classical.choice, Quot.sound]

'PrincipiaTractalis.SubstrateCompletionFaithful.
  substrate_completion_simple_unconditional'
  depends on axioms: [propext, Classical.choice, Quot.sound]

'PrincipiaTractalis.SubstrateTraceUniqueness.substrate_UHF_trace_unique'
  depends on axioms: [propext, Classical.choice, Quot.sound]
```

These are the three standard Lean 4 kernel axioms: propositional extensionality, choice, and quotient soundness. There are no project-level axioms and no `sorry` on any code path leading to the theorems of this paper. The development is classical, not constructive: choice enters through `mathlib`'s extension of uniformly continuous maps to a completion (`UniformSpace.Completion.extension`) and the universal properties of the direct limit, both used pervasively here. We make no claim of computable or choice-free content.

Content	Section	Lean file (under PF/)
Tower, limit, completion, instances	§2	<code>Substrate* chain</code>
Trace and soft properties	§3	<code>SubstrateUHFTraceIs*.lean</code>
Cauchy–Schwarz for the trace	§4.1	<code>SubstrateUHFTraceCauchySchwarz.lean</code>
Trace-null ideal; faithful-of-simple	§4.2	<code>SubstrateUHFTraceNullIdeal.lean</code>
Weyl averaging	§4.3	<code>SubstrateMatrixUnitaryAveraging.lean</code>
Direct-limit simplicity	§4.4	<code>SubstrateDirectLimitSimplicity.lean</code>
Four-fold equivalence	§4.5	<code>SubstrateTraceDetectionAttempt.lean</code>
Level expectation E_k	§5.1	<code>SubstrateConditionalExpectation.lean</code>
2-norm contraction	§5.2	<code>SubstrateCondExpContraction.lean</code>
Operator-norm contraction	§5.3	<code>SubstrateCondExpOpNorm.lean</code>
Extension; faithfulness; simplicity	§5.4	<code>SubstrateCompletionFaithful.lean</code>
Uniqueness; factor capstone	§5.5	<code>SubstrateTraceUniqueness.lean</code>

TABLE 1. Section-to-source map.

7. RELATED WORK

`Mathlib`'s C^* -algebra hierarchy, continuous functional calculus, matrix positivity infrastructure, direct limits, and completions, all community developments with functional-analysis foundations described in [4], are the platform this work builds on. To the best of our knowledge, after searching `mathlib`, the Coq `opam` package archive, and the Isabelle Archive of Formal Proofs for UHF, AF, Glimm, simple- C^* , and tracial-state content, and after consulting the operator-algebra formalization literature, no prior formalization in Lean, Coq, or Isabelle contains a concrete infinite-dimensional simple C^* -algebra, a canonical faithful tracial state on one, or any version of Glimm's simplicity theorem [5]. We have not performed an exhaustive survey of every proof-assistant library and would welcome pointers to prior art.

On the classical side, everything proved here is textbook material [3, 1, 12]: Glimm's classification and simplicity [5], Powers' work on the associated factors [10], Bratteli's AF framework [2], and the conditional-expectation theory of Tomiyama [13], descending from Kadison [6] and Stinespring [11].

The contribution is not new mathematics about M_{3^∞} ; it is the first machine-checked path to these facts, together with the elementary replacement proofs of Sections 5.2 and 5.3 that the setting demanded.

Several extensions are natural. The three contributed files can be upstreamed, and the tower can be de-specialized from base 3 to an arbitrary supernatural number, since most proofs are already base-generic in structure. A general complete-positivity API, with partial traces and Kadison–Schwarz, now has a worked concrete example to test against. The GNS representation of τ_{UHF} and the trace on the double commutant would connect this construction to the hyperfinite II_1 factor [9, 10], and Glimm’s classification by supernatural number is a further target for which the clopen spectral projection API is a first ingredient.

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The Lean 4 and mathlib community built the C^* -algebra, direct-limit, completion, and functional-calculus infrastructure on which every theorem here stands.

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MESA, ARIZONA, USA

Email address: psolorzano@gmail.com

URL: ORCID:~0009-0002-0734-5565